Homework #4: X-ray diffraction

Due date: Oct. 26th 2025

Structure factor for a tri-atomic basis

1. Consider a linear crystal with lattice constant a in which the basis consists of atom of species A at 0 and two atoms of species B at 1/3 and 2/3. Find the structure factor of such a crystal. X-rays of wavelength λ $(= a/4)$ irradiate this crystal at an oblique incidence (45°). Using the Ewald construction in the plane of incidence, show graphically, the diffraction directions and indicate the relative weights of the different diffracted intensities. State the physical significance of the vector $\boldsymbol{k}$ from which the extremity is on a plane passing through the origin of the reciprocal lattice.

2. The main experimental methods of X-ray diffraction are:
   (a) the Laue method
   (b) the crystal rotation method
   (c) the powder method (or the Debye–Scherrer method)
   Describe the corresponding experimental set-up and with the help of the Ewald construction in reciprocal space, deduce the observed diffraction patterns in each case.